

Module : STM

Exercice 1: (10 pts)

1) a) $x = (45,5)_{10} = (101101,1)_2$ (2)

- partie entière : $(45)_{10} = (101101)_2$
- " fractionnaire : $0,5 \times 2 = 1,0$ donc $0,5 = (0,1)_2$

b) $y = (14,75)_{10} = (1110,11)_2$ (2)

- partie entière : $(14)_{10} = (1110)_2$
- partie fractionnaire : $0,75 \times 2 = 1,5$ (1, reste 0,5)
 $0,5 \times 2 = 1,0$ (1, reste 0) donc $0,75 = (0,11)_2$

2) Format IEEE 754 16bits

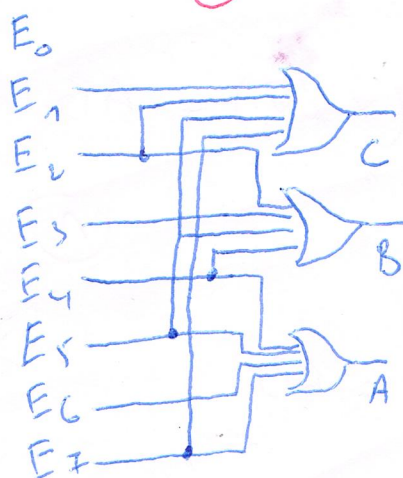
- 1 bit de signe
- 5 bits pour l'exposant (avec biais de 15)
- 10 bits pour la mantisse (sans le 1 explicite)

$$x = 0 \ 10100 \ 0110110000$$
 (2)

$$y = 0 \ 10010 \ 1101100000$$
 (2)

3) Encoder 3 bits: Décimal $\rightarrow$ Gray

E_i	A	B	C
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1



$$A = 4 + 5 + 6 + 7$$

$$B = 2 + 3 + 4 + 5$$

$$C = 1 + 2 + 5 + 7$$

4) Transcoder Gray $\rightarrow$ X53

A	B	C	X	Y	Z
0	0	0	0	1	1
0	0	1	1	0	0
0	1	0	1	1	0
0	1	1	1	0	1
1	0	0	x	x	x
1	0	1	x	x	x
1	1	0	1	1	1
1	1	1	x	x	x

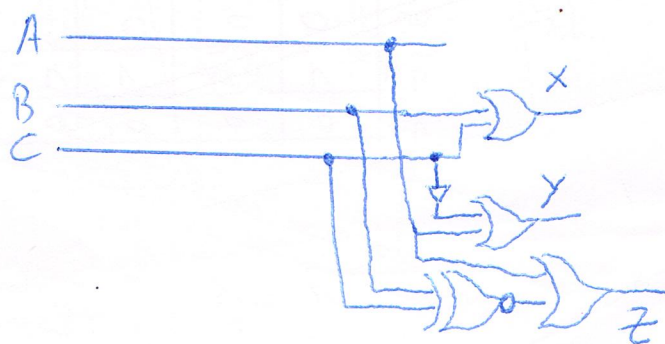
ABC	00	01	11	10
0	0	1	1	1
1	x	x	x	1

$$x = B + C$$

ABC	00	01	11	10
0	1	0	0	1
1	x	x	x	1

$$y = \bar{C} + A$$

ABC	00	01	11	10
0	1	0	1	0
1	x	x	x	1

$$z = A + \bar{B}\bar{C} + BC = A + B \oplus C$$


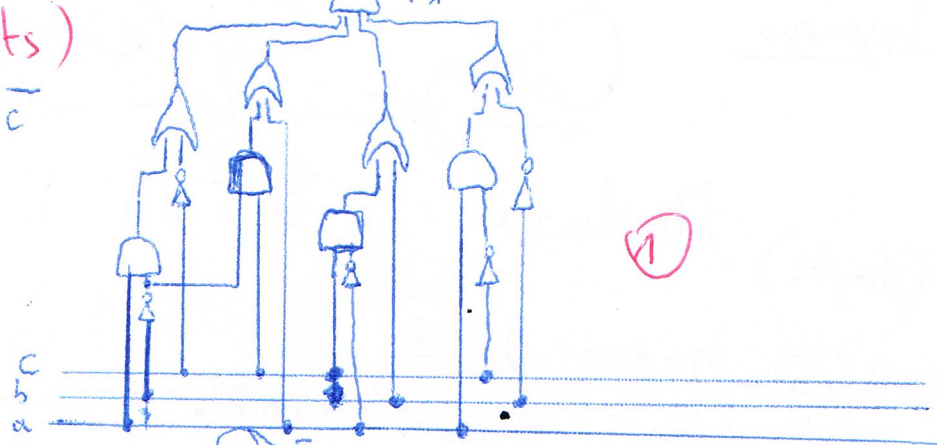
(1)

Exercice N°2: (06 pts)

1) a b c $F_1 = abc$

a	b	c	F_1
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

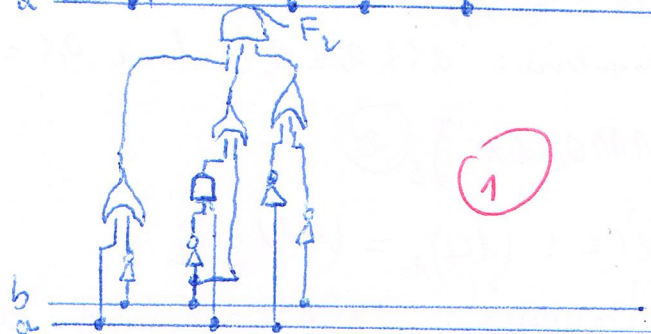
(1)



(1)

a	b	F_2
0	0	0
0	1	0
1	0	1
1	1	0

(1)



(1)

2) $(\overline{A}B) \oplus (\overline{A}B) \oplus (\overline{A}B)$

A	B	$\overline{A}B$	$\overline{A}B$	$\overline{A}B$	F_1
0	0	1	0	1	0
0	1	1	1	1	0
1	0	1	0	0	1
1	1	0	0	1	1

(1)

$$3) F_1 = (AB\overline{C}\overline{X}) + (AB\overline{C}X) + (ABC\overline{X}) + (ABCX)$$

$$= AB(\overline{C}\overline{X} + \overline{C}X + C\overline{X} + CX)$$

$$= AB(X(\overline{C} + C) + \overline{X}(\overline{C} + C)) = AB(X + \overline{X}) = AB$$

(1)

$AB \backslash CX$	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	0	0	0	0

$$F_1 = AB$$

(2)

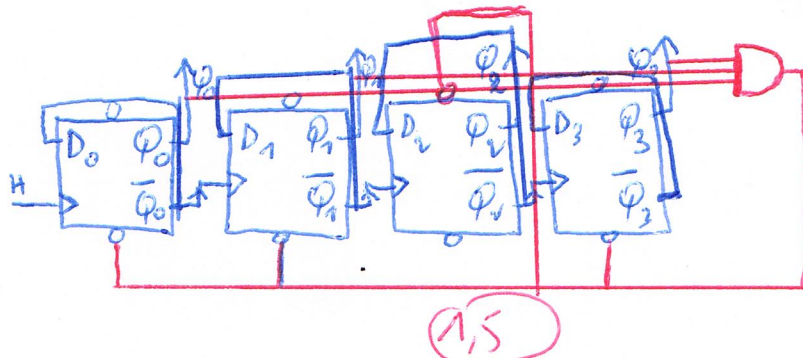
Exercice 03: (04 pts)

1)

N	Q_0	Q_1	Q_2	Q_3
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

0,5

à la place de 12
ajoutant 2
 $Q_3 \cdot Q_1 \cdot Q_0$



1,5

2)

0,5

$Q_a \backslash Q_b$	0	1
0	0	1
1	1	0

$$D_b = \overline{Q_a} \cdot Q_b + Q_a \cdot \overline{Q_b} \\ = Q_a \oplus Q_b$$

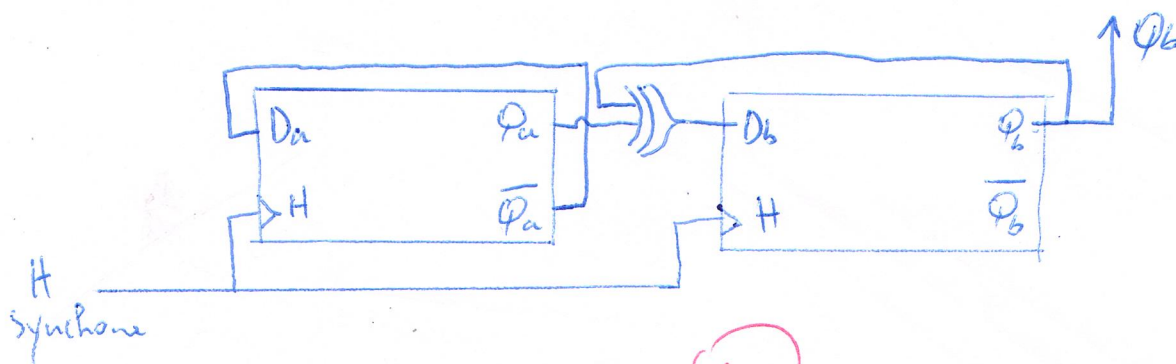
0,5

$Q_a \backslash Q_b$	0	1
0	1	1
1	0	0

$$D_a = \overline{Q_a}$$

Nbre	Q_b, Q_a	D_b	D_a
0	0 0	0	1
1	0 1	1	0
2	1 0	1	1
3	1 1	0	0

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1



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